

Class 12: Genome Informatics

Brian Wong (PID: A18639001)

Background

This class introduces the fundamentals of genome informatics and high throughput sequencing. The lab itself focuses on the Galaxy platform and RNA-Sequencing methods for gene expression analysis.

Section 4: Population Scale Analysis [HOMEWORK]

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

The sample size for each genotype is here below:

```
table(expr$geno)
```

```
A/A A/G G/G  
108 233 121
```

The median expression levels for each of the genotypes are here below:

```
summary(expr$exp[expr$geno == "A/A"])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
11.40	27.02	31.25	31.82	35.92	51.52

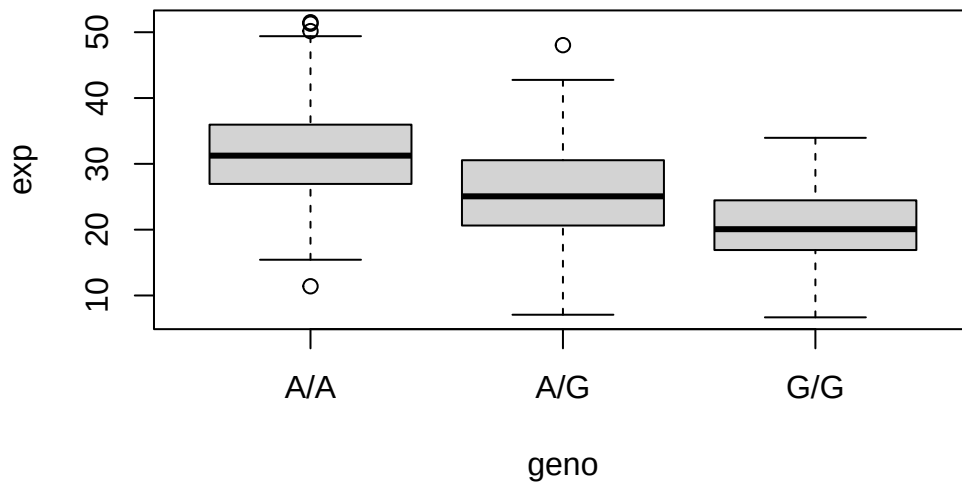
```
summary(expr$exp[expr$geno == "A/G"])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
7.075	20.626	25.065	25.397	30.552	48.034

```
summary(expr$exp[expr$geno == "G/G"])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
6.675	16.903	20.074	20.594	24.457	33.956

```
bp <- boxplot(exp ~ geno, expr)
```



```
bp$stats[3,]
```

```
[1] 31.24847 25.06486 20.07363
```

A/A = 31.24847, A/G = 25.06486, G/G = 20.07363

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP affect the expression of ORMDL3?

It can be inferred that the relative expression value between A/A and G/G are quite different, with G/G being having a reduced relative expression level compared to A/A. Therefore, it can be inferred that the SNP does affect the expression of ORMDL3 based on this boxplot.

```
library(ggplot2)
ggplot(expr) + aes(geno, exp, fill = geno) + geom_boxplot(notch = TRUE)
```

